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$\mathcal{M}$ INFINITE DIMENSION

$B(\mathcal{M})$ IS ITSELF A VECTOR SPACE, SINCE THE LINEAR COMBINATION OF TWO OPERATORS IS ALSO AN OPERATOR
 ↳ SET OF ALL OPERATORS OF $\mathcal{M}$

INNER PRODUCT

$$\langle A, B \rangle = \text{Tr}(A^\dagger B)$$

TRACE IS A PROPERTY OF AN OPERATOR

- $\langle A, A \rangle = \langle A, A \rangle^*$
- $\langle A, A \rangle \geq 0$

REMEMBER:

$$\langle \cdot \rangle \rightarrow \mathbb{C}$$

$$(A^\dagger B)^\dagger = (B^\dagger A)^\dagger$$

$$\begin{bmatrix} A^{\dagger\dagger} = A \\ A^{\dagger\dagger\dagger} = A^\dagger \end{bmatrix}$$

↳ BECAUSE

$$(B^\dagger A)^\dagger = (B^{\dagger\dagger} A)^{\dagger\dagger} = (A^\dagger (B^{\dagger\dagger})^\dagger)^\dagger = A^{\dagger\dagger} B = A^\dagger B$$

$$\rightarrow 1) \langle A, B \rangle = \text{Tr}(A^\dagger B) = \text{Tr}(B^\dagger A)^\dagger = (\text{Tr}(B^\dagger A))^* = \langle B, A \rangle^*$$

$$\rightarrow 2) \langle A, A \rangle = \text{Tr}(A^\dagger A) = \langle A^\dagger A \rangle = \langle \psi | A^\dagger A | \psi \rangle = \langle \psi^\dagger | \psi^\dagger \rangle \geq 0$$

LET'S CONSIDER A QUANTUM SYSTEM

THE PAULI OPERATOR FORM THE SO CALLED PAULI BASIS $\{\hat{x}, \hat{y}, \hat{z}\}$

$$\hat{x} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \hat{y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\mathcal{B} = \{|\hat{x}\rangle, |\hat{y}\rangle\}$$

REMEMBER $\langle A, B \rangle = \text{Tr}(A^\dagger B)$

$$\hat{x}^2 = \hat{y}^2 = \hat{z}^2 = \hat{1} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{1}$$

$$\|A\| = \sqrt{\langle A, A \rangle}$$

$$\hat{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\langle \hat{1}, \hat{1} \rangle = \text{Tr}(\hat{1} \hat{1}) = \text{Tr}(\hat{1}) = 2$$

$$\text{Tr} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = A + D$$

$$\bullet \langle \hat{x}, \hat{x} \rangle = \sqrt{2} = \langle \hat{y}, \hat{y} \rangle = \langle \hat{z}, \hat{z} \rangle$$

$$\bullet \langle \hat{x}, \hat{y} \rangle = \langle \hat{y}, \hat{z} \rangle = \langle \hat{z}, \hat{x} \rangle = \langle \hat{1}, \hat{x} \rangle = \langle \hat{1}, \hat{y} \rangle = \langle \hat{1}, \hat{z} \rangle = 0$$

↓

$$\langle \hat{x}, \hat{y} \rangle = \text{Tr}(\hat{x}^\dagger \hat{y})$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \rightarrow \text{Tr} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = 0$$

$$\langle \gamma, \beta \rangle = \text{Tr}[\gamma^\dagger \beta] = 0 \quad \gamma^\dagger = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

THE PAULI OPERATORS FORM THE SO-CALLED **PAULI BASIS** $\{I, X, Y, Z\}$

$$\rho_{\text{qubit}} = \frac{1}{2} \left(I + a_x X + a_y Y + a_z Z \right) \quad a_x, a_y, a_z \in \mathbb{R}$$

• The Pauli operators form the so-called **Pauli basis**. $\{I, X, Y, Z\}$

$$\rho_{\text{qubit}} = \frac{1}{2} (I + a_x X + a_y Y + a_z Z) \quad a_x, a_y, a_z \in \mathbb{R}$$

$$\rho_{\text{qubit}} = \frac{1}{2} \left(I + a_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + a_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + a_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)$$

$$= \frac{1}{2} \begin{pmatrix} 1 + a_z & a_x - ia_y \\ a_x + ia_y & 1 - a_z \end{pmatrix}$$

$$\rho_{\text{qubit}} = \frac{1}{2} (I + a_x X + a_y Y + a_z Z) \quad a_x, a_y, a_z \in \mathbb{R}$$

$$\text{Formally, } \rho_{\text{qubit}} = \frac{1}{2} (I + \mathbf{a} \cdot \boldsymbol{\sigma}) \quad \boldsymbol{\sigma} = (X, Y, Z)$$

$$\mathbf{a} = (a_x, a_y, a_z) \quad \text{Bloch vector}$$

$$\rho_{\text{qubit}} = \frac{1}{2} (I + a_x X + a_y Y + a_z Z) \quad \text{ANY REAL Bloch vector } \mathbf{a} \text{ defines a trace-1 HERMITIAN OPERATOR}$$

